

Myxospores isolation individual FBs protocols

Link to the protocol

Protocol: Myxospores isolation

- Take pictures of FBs plates before and after harvest
- Harvest 2 times 20 FBs for each experimental conditions in:
 - 100 μ L BSA (*200 concentration condition*)
 - 1 mL BSA (*'Usual' condition*)
- 2 washing: centrifugation 10 min at $8000 \times g$ with MM,
- Complete volume to 200 μ L (*200 concentration condition*) and 2 mL (*Usual condition*),
- Sonication: *Time* 2 min, *Pulse* 10 sec ON / 15 sec OFF, *Amplitude* 28%, *Probe* 60-62k,
- Heat Treatment 2 h,
- Only for Usual condition: Concentration at 200 μ L then homogenisation (2nd sonication) *Time* 30 sec at same parameters (c.f. 1st sonication parameters above),
- Imaging at $\times 40$ in phase contrast (and fluorescence) using Malassez slide (counting chamber)

Protocol: Image processing

BE CAREFUL Images must be taken off grid in the counting chamber. Check script/notebook, there is conditions to satisfy before image processing.

- Prepare three working directories:
 - 1-Input_images;
 - 2-Phase_contrast_images;
 - 3-Segmented_images
- On Fiji: Execute the macro channel splitter [Macro_Fildir_channel_splitter.ijm](#)
- On Fiji: Execute **Plungins** > **Labkit** > **Macro Recordable** > **Segment Images in Directory with Labkit** with the classifier file [Classifier_sp_vc.classifier](#)
- Execute [Automated_workflow.py](#)
- A new directory would appear with the Result dataset